

Alison Seongjee Park

Bioinformatician | Computational Oncology | Multi-Omics & Single-Cell Analysis

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Bioinformatician at Icahn School of Medicine at Mount Sinai applying multi-omics integration, single-cell RNA-seq analysis, and unsupervised learning to multiple myeloma. Contributed to 8 verified public outputs, including a 2026 *Journal of Medicinal Chemistry* article and second-author abstracts spanning 389-patient and 776,859-cell studies.

EXPERIENCE

Bioinformatician

Dec 2023–Present

Icahn School of Medicine at Mount Sinai

Multiple Myeloma Computational Research

- Integrated bulk RNA-seq, single-cell RNA-seq, and clinical data using SNF, cNMF, MOVICS, MOFA, and integrAO to stratify newly diagnosed multiple-myeloma cohorts.
- Built patient-similarity network analyses that refined disease classification beyond traditional staging and surfaced immune-based subtypes with distinct outcomes.
- Performed Scanpy and pySCENIC workflows that informed second-author studies spanning 389 patients and 776,859 cells.

Research Assistant

Aug 2018–Nov 2021

Icahn School of Medicine at Mount Sinai

Cancer Epigenetics Laboratory

- Studied SETD8 and E2F1 in multiple myeloma and mantle cell lymphoma using in vitro and in vivo models, contributing to ASH abstracts and later medicinal-chemistry publication.
- Analyzed experimental datasets with R, Prism, and survival analysis while working in cross-functional translational oncology teams.

Earlier Research Training

2015–2018

Mount Sinai and Northwell Health

- Built cancer-biology foundations through internships in chronic lymphocytic leukemia and chemo-resistant bladder cancer, developing wet-lab familiarity with cell culture, flow cytometry, immunohistochemistry, PCR, genotyping, and mouse-model workflows.

EDUCATION

Sophie Davis / The City College of New York

BS Biomedical Sciences, Cum Laude | 2021

Flatiron School

Data Science Bootcamp | 2023

CUNY School of Medicine

Medical training period | 2021–2023

TECHNICAL SKILLS

Programming:

Python, R, SQL, shell scripting, Git/GitHub

ML / Statistics:

scikit-learn, survival analysis, clustering, dimensionality reduction, feature engineering

Genomics:

Scanpy, Seurat, pySCENIC, bulk RNA-seq, single-cell RNA-seq, DESeq2, GSEA

Infrastructure:

Docker, HPC, reproducible workflows, multi-institutional dataset management

SELECTED RESEARCH OUTPUTS

2026.

Discovery of a Potent, Selective, and In Vivo Efficacious Covalent Inhibitor for Lysine Methyltransferase SETD8. *Journal of Medicinal Chemistry*.

2025.

Integrating Microenvironment with Tumor Multi-Omic using Unsupervised Machine Learning to Model Heterogeneity Refines Multiple Myeloma Subtypes and Reveals Immune-Based Clusters with Prognostic Impact. *Clinical Lymphoma, Myeloma and Leukemia*.

2024.

Immune Profiling Reveals Distinct Features of Multiple Myeloma Subtypes Defined By Multi-Omics Network Analysis. *Blood*.

AWARDS

ASH Abstract Achievement Award (2), 2024 • UKC FIRE 2025, 2nd place • Cum Laude, 2021